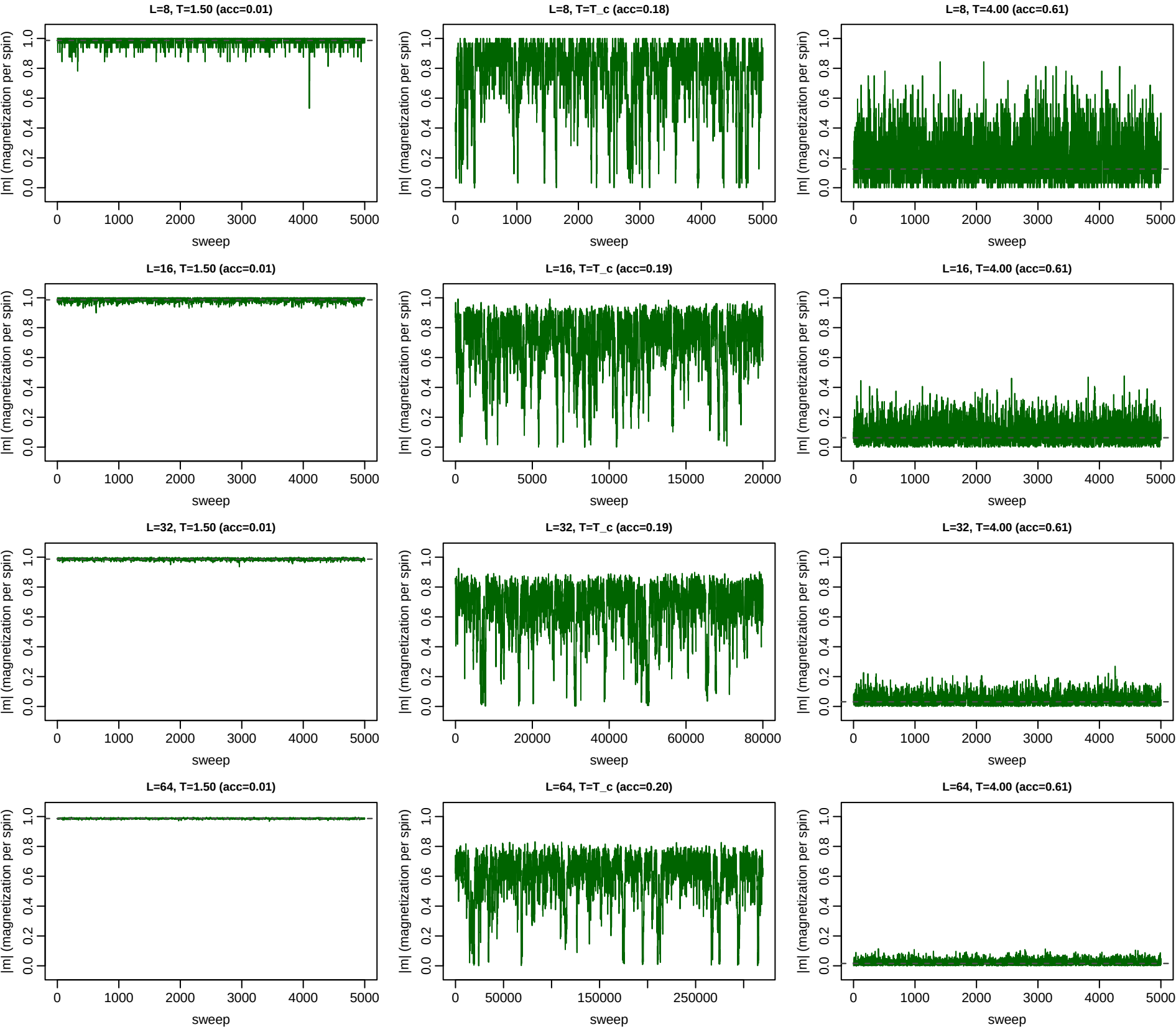Magnetization trace